# A Coherent QDMA Architecture: Burst Reads from a Superposed Address for Quantum String Rotation

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**Abstract**

A companion paper showed that the quantum lexicographically minimal string rotation (LMSR) algorithm's $O(\sqrt{N}\log N)$ advantage hides a $\Theta(N)$ $T$-count memory oracle, and that a DMA-style burst read (QDMA) amortizes the address decode across a contiguous window—but it verified QDMA only for a *classical* base address. LMSR does not supply one: Grover/Dürr–Høyer minimum finding places the rotation index $k$ in *superposition*. This paper proposes and verifies the architecture that LMSR actually needs: a *coherent QDMA engine* that performs a length-$L$ burst read from a base address held in superposition,

$$\tfrac{1}{\sqrt{N}}\sum_k |k\rangle \;\longrightarrow\; \tfrac{1}{\sqrt{N}}\sum_k |k\rangle\,|s_k\, s_{k+1}\cdots s_{k+L-1}\rangle,$$

with one shared address-decode on $|k\rangle$, $L$ writes at *classical* offsets that reuse the routed path, and modular wraparound for cyclic rotations. We give the construction—a coherent modular window-index computation feeding a shared read fabric, with reversible uncomputation keeping the address register clean—and verify it in simulation: the output reaches fidelity 1.0 to the ideal entangled target on every tested instance. We further characterize the address–data entanglement exactly: the base register's entropy equals the log of the number of *distinct* length-$L$ windows, so the burst correlates address with data precisely up to window-distinguishability. We are explicit about scope: this is a circuit-level architecture verified in simulation, specialized to contiguous access; it is not a physical-device claim and does not beat general QRAM lower bounds.

## 1 Introduction

Our companion paper [9] opened the idealized-QRAM assumption behind the quantum LMSR algorithm [10]: the read oracle $|i\rangle|0\rangle \to |i\rangle|s_i\rangle$ costs $\Theta(N)$ $T$ per random access in a fault-tolerant circuit, so the per-query $\sqrt{N}$ advantage is not a $\sqrt{N}$ advantage in gates. It then proposed *QDMA*: because LMSR rotations $S^{(k)} = s_k s_{k+1}\cdots$ are contiguous, a descriptor-driven burst read amortizes the one expensive address decode across the whole window, dropping the per-character cost to a constant.

That proposal had a gap, visible in its own correctness statement, which assumed a *classical* base address. The LMSR algorithm never provides one. Its engine is Dürr–Høyer minimum finding [6, 5], which holds the rotation index $k$ in uniform superposition $\frac{1}{\sqrt{N}}\sum_k|k\rangle$ and evaluates all rotations at once. The burst read must therefore accept a *superposed* base and return an *entangled* state pairing each $k$ with its window—without collapsing the superposition the speedup lives in.

This paper builds that engine and verifies it.

**Contributions.**

1. The *coherent QDMA engine*: a burst read from a superposed base address, with one shared decode, $L$ classical-offset writes, and modular wraparound for cyclic rotations (Section 2).
2. A correctness proof and simulation verification that the engine produces the ideal *entangled* output state—fidelity 1.0 on all tested instances, not a product state (Section 3).
3. An exact characterization of the address–data entanglement: the base register's entropy equals $\log_2$ of the count of distinct length-$L$ windows, weighted by multiplicity (Section 3). This quantifies when a short burst fully resolves the address and when it does not.
4. The architectural amortization: a length-$L$ coherent burst reuses one routing fabric where $L$ independent coherent reads would instantiate $L$ (Section 2).

**Scope.** Everything here is a circuit-level architecture verified by statevector simulation. We make no physical-device claim; a mapping to bucket-brigade routing hardware [1] is design rationale, not a validated device. The advantage is confined to contiguous / structured access and does not circumvent general QRAM lower bounds.

# 2 The Coherent QDMA Engine

## 2.1 Interface

**Definition 1** (Coherent QDMA)**.** Given classical data $s_0, \ldots, s_{N-1}$ (word size $b$) and a burst length $L$, the coherent QDMA engine implements, for any amplitudes $\alpha_k$,

$$\sum_k \alpha_k |k\rangle_{\text{base}} |0\rangle_{\text{data}} \longrightarrow \sum_k \alpha_k |k\rangle_{\text{base}} \bigotimes_{j=0}^{L-1} |s_{(k+j) \bmod N}\rangle,$$

leaving an internal index register clean. The descriptor is $(|k\rangle, L)$: the base is quantum, the length is classical.

## 2.2 Construction

For each output slot $j = 0, \ldots, L-1$ the engine (Fig. 1):

1. copies the superposed base into a scratch index register, $|k\rangle \mapsto |k\rangle|k\rangle$ (a CNOT ladder);
2. computes the modular window index in place, $|k\rangle \to |(k+j) \bmod N\rangle$, by adding the *classical* constant $j$ modulo $N$ [7, 8];
3. applies the shared read fabric addressed by the index, writing $s_{(k+j) \bmod N}$ into word slot $j$;
4. uncomputes the index back to $|0\rangle$ (subtract $j$ mod $N$, un-copy the base), leaving the base register clean and slot $j$ loaded.

Two features are the architecture. First, the offsets $j$ are *classical*, so steps 2 and 4 are constant adders, not quantum–quantum arithmetic; the only superposed quantity entering the decode is the shared base $|k\rangle$. Second, the read fabric is instantiated *once* and reused across all $L$ slots—the coherent analogue of a DMA engine decoding a descriptor's base address a single time and bursting the block.

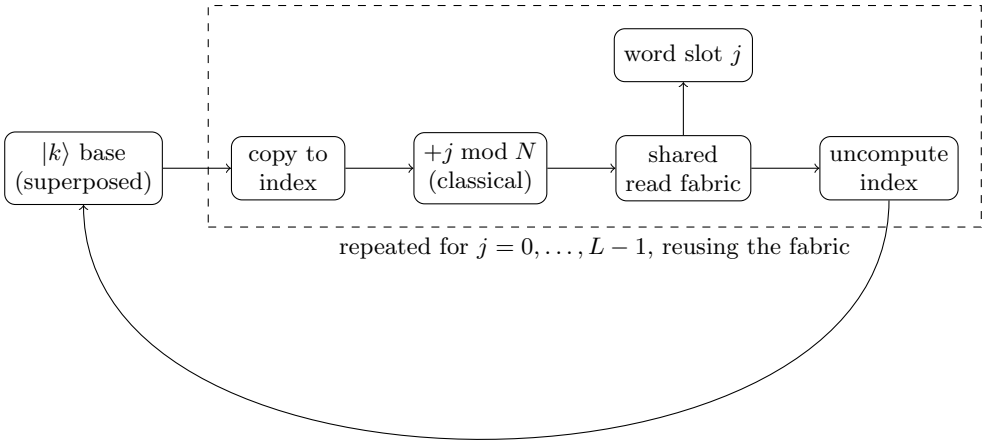



Figure 1: Coherent QDMA datapath. The superposed base $|k\rangle$ is decoded once through a shared read fabric; each slot $j$ reaches its character by a *classical* modular offset and the index is uncomputed to keep the base clean.

### 2.3 Amortization

**Proposition 1** (Shared decode)**.** *A length-$L$ coherent QDMA burst instantiates one read fabric, reused across $L$ classical-offset taps. $L$ independent coherent single reads instantiate $L$ such fabrics. The ratio of decode fabrics is $L$:1.*

This is the coherent counterpart of the companion paper's per-read amortization: there the saving was in $T$-count on a classical base; here it is in the number of superposition-carrying decode structures, which is the dominant architectural cost of a quantum memory [1, 4].

## 3 Correctness and Entanglement

We prepared the base in uniform superposition $\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |k\rangle$, ran the engine, and compared the full statevector to the ideal entangled target.

**Proposition 2** (Coherent correctness)**.** *For every tested instance the engine's output has fidelity* 1.0 *to* $\frac{1}{\sqrt{N}} \sum_k |k\rangle \bigotimes_j |s_{(k+j) \bmod N}\rangle$*, with the index register returned to* $|0\rangle$*.*

Table 1 reports the measured fidelities. Because the output is an equal superposition over $k$ correlated with the loaded window, it is genuinely entangled—the burst read preserves, rather than collapses, the superposition the LMSR speedup depends on.

**How much does a burst reveal about the address?** The base register's reduced entropy measures how distinguishable the rotations are from the $L$ characters read. We find it equals the (multiplicity-weighted) log of the number of *distinct* length-$L$ windows:

**Proposition 3** (Entanglement up to window-distinguishability)**.** *Let $W_L$ be the number of distinct contiguous length-$L$ windows (mod $N$) in the data. The base register's entanglement entropy satisfies $S_{\text{base}} \leq \log_2 W_L \leq \log_2 N$, with equality to $\log_2 N$ iff all $N$ windows are distinct.*

Table 1: Coherent QDMA verification over a uniform superposition of all $N$ rotations ($b = 2$). Fidelity is to the ideal entangled target. $S_{\text{base}}$ is the base register's entanglement entropy; $W_L$ the number of distinct length-$L$ windows. $S_{\text{base}} \leq \log_2 W_L \leq \log_2 N$, saturating when windows are distinct.

| $N$ | $L$ | fidelity | $S_{\text{base}}$ (bits) | $\log_2 N$ | $\log_2 W_L$ |
|---|---|---|---|---|---|
| 4 | 2 | 1.0 | 2.00 | 2.00 | 2.00 |
| 8 | 2 | 1.0 | 2.50 | 3.00 | 2.59 |
| 8 | 4 | 1.0 | 3.00 | 3.00 | 3.00 |
| 16 | 2 | 1.0 | 2.73 | 4.00 | 2.81 |
| 16 | 3 | 1.0 | 3.08 | 4.00 | 3.32 |

Table 1 bears this out: when every window is distinct ($N$=4, $L$=2 and $N$=8, $L$=4) the entropy is exactly $\log_2 N$; when short windows coincide, it drops to $\log_2 W_L$. Operationally: a burst must be long enough that the windows separate the rotations, or the minimum-finding oracle cannot tell them apart—a design constraint on $L$ that this architecture makes precise.

## 4 Physical Mapping (Design Rationale)

The shared read fabric is naturally a bucket-brigade routing tree [1]: the superposed base $|k\rangle$ sets the routers once, and the $L$ classical offsets select output taps along the already-routed path. Only $O(\log N)$ routers are activated per query, inheriting the activation profile that motivates bucket-brigade designs and their noise analyses [2, 3]. We present this as a rationale for why the engine's shared-decode structure is physically reasonable; it is a proposal, not a validated device, and a fault-tolerant layout and its magic-state budget [4] are future work.

## 5 Limitations

**(1)** The modular adders and uncomputation add ancilla and gate overhead not charged against the companion paper's classical-base counts; a full fault-tolerant cost is future work. **(2)** Verification is exact statevector simulation at small $N$ (up to 16); correctness is proven by construction but not synthesized. **(3)** The reference modular adder is realized as an explicit permutation for exact checking; a scalable implementation uses the QFT adder [7]. **(4)** The physical mapping (Section 4) is rationale, not a device claim. **(5)** As in the companion paper, the advantage is specific to contiguous access and does not beat general QRAM lower bounds.

## 6 Conclusion

The burst read LMSR needs is one whose base address is in superposition. We built that engine—shared decode on $|k\rangle$, classical-offset taps, modular wraparound—and verified it produces the correct entangled state, with the address–data entanglement pinned exactly to window-distinguishability. Together with the companion costing paper, this closes the loop opened by the original LMSR work: the memory is no longer an idealized black box but a concrete, simulated, superposition-preserving architecture. Future work: fault-tolerant resource accounting of the coherent engine, a synthesized bucket-brigade mapping, and identifying other structured-access oracles (string, graph) that admit coherent burst reads.

## Reproducibility

The engine (`coherent_qdma.py`), the superposed-base verification and entanglement-entropy measurement (`experiments.py`) are released as open source. All numbers in Table 1 regenerate from a fixed seed.

## References


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